Derivation of Sparse NMF with KL solved by IADMM (Paper Style)

Step 1: NMF formulation. Given the observed spectrogram

$$V \in \mathbb{R}_{\geq 0}^{F \times T},$$

approximate it as

$$V \approx WH, \tag{1}$$

where $W \in \mathbb{R}_{\geq 0}^{F \times R}$ and $H \in \mathbb{R}_{\geq 0}^{R \times T}$.

Step 2: Objective with KL divergence and sparsity. The optimization problem is

$$\min_{W,H \geq 0} \; D_{\mathrm{KL}}(V \| WH) + \lambda \|H\|_1, \tag{3–4}$$

with

$$D_{\mathrm{KL}}(V \| X) = \sum_{i,j} \Big(V_{ij} \log \frac{V_{ij}}{X_{ij}} - V_{ij} + X_{ij} \Big).$$

Step 3: Variable split for ADMM. Introduce an auxiliary variable X:

$$\min_{X,W,H \geq 0} \; D_{\mathrm{KL}}(V \| X) + \lambda \|H\|_1 \quad \text{s.t. } X = WH. \tag{5}$$

Step 4: X–update. The subproblem is

$$\min_{x \geq 0} \; v \log \frac{v}{x} - v + x + \tfrac{\rho}{2}(x - z)^2,$$

with $v = V_{ij},\ z = (WH - \alpha)_{ij}$. Derivative condition:

$$\rho x^2 + (1 - \rho z)x - v = 0. \tag{7}$$

Positive solution:

$$x_{ij}^+ = \frac{(\rho z_{ij} - 1) + \sqrt{(\rho z_{ij} - 1)^2 + 4\rho v_{ij}}}{2\rho}. \tag{8}$$

Step 5: W–update. Solve a least squares system with its block dual α_W:

$$(HH^\top)\, W^+ = (X^+ + \alpha_W)H^\top, \qquad W^+ \leftarrow \max(W^+, 0). \tag{1}$$

Step 6: H–update. With block dual α_H:

$$(W^{+\top} W^+)\, \widetilde{H} = W^{+\top}(X^+ + \alpha_H), \tag{2}$$

then apply sparsity and nonnegativity:

$$H^+ = \max\Big(0,\ \widetilde{H} - \tfrac{\lambda}{\rho}\Big). \tag{3}$$

Step 7: Dual updates.

$$\alpha_W^+ = \alpha_W + (X^+ - W^+H^+), \qquad \alpha_H^+ = \alpha_H + (X^+ - W^+H^+). \tag{4}$$

Step 8: Ill-conditioning. The condition number is defined as

$$\mathrm{cond}(A) = \|A\| \, \|A^{-1}\|, \tag{9}$$

and a large value indicates an ill-conditioned system.

Step 9: IADMM stabilization via PEWI. For $AX = b$, define

$$\widetilde{A} = A + \varepsilon I, \quad \varepsilon > 0, \tag{10}$$

then use the iterative scheme

$$X^{k+1} = (\widetilde{A})^{-1}(b + \varepsilon X^k). \tag{11–12}$$

This improves conditioning and stabilizes the W and H updates.